

Validity Analysis: MMLU

Target context: South Asian Agricultural and Environmental Science —
Mymensingh–Telangana–Andhra Pradesh Deployment

Overall risk: **HIGH**

Deployment Context

Use case: Task: Environment science and geograpy-specific agricultural knowledge understanding

LLMs evaluated: Generic frontier model vs small llms as well as region specific llms

Domain: Environmental science

Setting: Environment science research utility assessment using llm and local-context

Target population: An environment scientist from Mymensingh, Bangladesh wants to get knowledge about different south asian region specific agricultural practice. So having different region and language specific agricultural and demographic context understanding (e.g. telegu spoken area, hydrabad India and agricultural practice/problems vs farmers practice/problems from Mymensingh, Bangladesh~bengali spoken in Mymensingh accented dialect.)

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	1	Serious concern	high
Input Content 	1	Serious concern	high
Input Form 	1	Serious concern	high
Output Ontology 	1	Serious concern	high
Output Content 	1	Serious concern	high
Output Form 	2	Significant gaps	medium
Average	1.2		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MMLU is fundamentally misaligned with the South Asian Agricultural and Environmental Science deployment across five of six dimensions. Five HIGH-priority dimensions (Input Ontology, Input Content, Input Form, Output Ontology, Output Content)

score 1 — the lowest possible — with multiple structural validity violations: complete absence of agro-ecological subjects, US/Anglophone-saturated content, English-only input, MCQ schema unable to encode regional pluralism, and US-anchored ground-truth labels with no South Asian annotator validation. Output Form scores 2, partially supporting cross-model comparison via accuracy metrics. The single biggest concern is that the benchmark cannot, by construction, measure the constructs the user needs to evaluate — there is no item in MMLU's 14,079-question test set addressing haor/wetland ecology, boro/aman/aus rice cultivation, char-land farming, Deccan dry-land cropping, or coastal aquaculture. Cross-border Bangladesh–India water-policy questions, which are actively contested in 2025, are entirely absent and could not be represented in the four-option MCQ format even if added. The benchmark is suitable only as a weak generic STEM-reasoning baseline for cross-model comparison.

Practical Guidance

What This Benchmark Measures

MMLU measures English-language MCQ accuracy on US/Western academic and professional knowledge across 57 subjects, with primary signal in mathematical reasoning, formal logic, and physical sciences (the dimensions where it scores least poorly for cultural neutrality). For this deployment, MMLU can provide a baseline cross-model accuracy comparison on culturally-neutral STEM and statistics items (per Strength 1 and Strength 3 in dataset analysis), and calibration diagnostics [Q60-62, Q106] useful for understanding model uncertainty behavior. It cannot measure South Asian agricultural or environmental science knowledge, nor any multilingual capability.

Construct Depth

Construct depth for the deployment's intended construct (South Asian agricultural and environmental science knowledge) is essentially zero: Input Ontology contains no agro-ecological subjects, Input Content contains no regional sourcing, Input Form precludes regional languages, Output Ontology cannot encode regional divergence, and Output Content has no regional annotator validation. Where MMLU does probe deeply (US legal/medical/historical knowledge), the construct is irrelevant to the deployment. The only modest depth is in language-independent STEM reasoning, which provides shallow but stable signal for cross-model comparison.

What Else You Need

Substantial supplementation is required across five dimensions: (1) Input Ontology — construct or adopt a South Asian agricultural knowledge taxonomy (BRRI/BARI/ICAR-aligned); (2) Input Content — recruit BAU, PJTSAU, ANGRAU agronomists to author items in Bengali, Telugu, and English covering haor/delta/dry-land/aquaculture systems; (3) Input Form — build multilingual evaluation infrastructure leveraging IndicTrans2 [WEB-18] and IndicBERT v2 [WEB-25] foundations, with explicit Bangladeshi-Bengali and Mymensingh-dialect coverage; (4) Output Ontology — design schema permitting regional-pluralism annotation for cross-border policy items; (5) Output Content — establish a regional annotator pool spanning Bangladesh and the Indian states, with explicit protocol for nationally-divergent answer keys (informed by current Ganges/Teesta context [WEB-1, WEB-2, WEB-4]). The synthetic agricultural QA methodology in arXiv 2507.16974 [WEB-26] provides a methodological precedent but covers only Hindi/Punjabi.

ID	Evidence
Q60	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)
Q106	"While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is $r = 0.81$, compared to $r = 0.63$ in the zero-shot setting." (p.13)
WEB-1	Active Bangladesh–India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)
WEB-2	Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-4	Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)

WEB-18	IndicTrans2 covers 22 Indian languages including Telugu but is a translation system, not knowledge benchmark (https://openreview.net/forum?id=vfT4YuzAYA)
WEB-25	IndicXTREME (ACL 2023) covers Bengali/Telugu for general NLU but no agricultural evaluation (https://aclanthology.org/2023.acl-long.693)
WEB-26	https://arxiv.org/html/2507.16974v2

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU’s 57-subject taxonomy is organized around US/Western academic disciplines [\[Q1, Q9, Q18\]](#) and contains no subject addressing agro-ecology, delta hydrology, haor wetland systems, boro/aman/aus rice cultivation, char-land farming, Deccan dry-land cropping, or coastal aquaculture — the precise subdomains required for this deployment. The dataset analysis confirmed this empirically across 175+ examples sampled from 37 configs: the closest environmental content observed is generic water chemistry and pastoralism definitions, with no South Asian agricultural systems represented [\[DATASET-D19, D22, D32\]](#). Many subjects are actively irrelevant (US foreign policy, US history, US government), introducing construct-irrelevant variance for an agricultural science deployment. Given IO is a HIGH-priority dimension and the gap is structural and complete, this is a major validity violation.

ID	Evidence
Q1	“We propose a new test to measure a text model’s multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more.” (p.1)
Q9	“The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more. It ranges in difficulty from an elementary level to an advanced professional level, and it tests both world knowledge and problem solving ability.” (p.1)
Q18	“We create a massive multitask test consisting of multiple-choice questions from various branches of knowledge. The test spans subjects in the humanities, social sciences, hard sciences, and other areas that are important for some people to learn. There are 57 tasks in total, which is also the number of Atari games (Bellemare et al., 2013), all of which are listed in Appendix B.” (p.3)
D19	The way of life based on breeding and herding of animals that is used as a source of food, shelter, and clothing is called pastoralism — Generic geography — no delta/wetland/aquaculture content
D22	In fungi, cell walls are reinforced by which of the following? (A) Chitin (B) Silica (C) Lignin (D) Collagen fibers — General biology — not agricultural/agro-ecological
D32	A substance that is NOT generally considered to be a toxic pollutant in water is (A) carbonic acid (B) a halogenated hydrocarbon (C) lead (D) mercury — Closest approach to environmental science — but generic water chemistry, not delta/haor/saline intrusion relevant

Strengths

- Universal STEM reasoning items (abstract algebra, formal logic, physics) provide a culturally-neutral baseline for cross-model comparison [\[DATASET Strength 1\]](#)
- Statistical inference items (hypothesis testing, sampling, Type I/II error) test methodology skills transferable to environmental science research [\[DATASET-D30\]](#)

ID	Evidence
D30	The Department of Health plans to test the lead level in a specific park. Because a high lead level is harmful to children, the park will be closed if the lead level exceeds the allowed limit. — Environmental testing question, but no South Asian or agricultural context

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for this deployment include haor/beel wetland ecology, boro/aman/aus rice cultivation, char-land farming, Brahmaputra-Jamuna floodplain hydrology, Deccan Rabi/Kharif dry-land cropping, brackishwater shrimp aquaculture, and Bangladesh-India transboundary water management — none of which are present in the 57-subject taxonomy.

IO-2: MMLU's taxonomy omits every regionally-relevant agricultural and environmental science category. Subjects are bounded by US/Western academic disciplines [Q9, Q18, Q33], with history/geography framed through Western lenses [Q28, DATASET-D16, D20]. The 'breadth' claim of 57 tasks [Q11, Q83] does not translate into agro-ecological coverage.

IO-3: MMLU includes substantial content irrelevant to South Asian agriculture: US Foreign Policy [Q171, DATASET-D5, D6], US History [Q151, DATASET-D2], US Government [DATASET-D3, D4], US tort/contract law [DATASET-D7, D8, D25], and Western European history [DATASET-D16, D17]. These consume evaluation weight without informing agricultural reasoning.

IO-4: The complete absence of any agro-ecological subject combined with heavy weighting toward US-centric subjects creates content underrepresentation (missing constructs) and construct-irrelevant variance simultaneously — both validity-harming patterns.

ID	Evidence
Q1	"We propose a new test to measure a text model's multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more." (p.1)
Q9	"The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more. It ranges in difficulty from an elementary level to an advanced professional level, and it tests both world knowledge and problem solving ability." (p.1)
Q11	"Since our test consists in 57 tasks, it can be used to analyze aggregate properties of models across tasks and to track important shortcomings." (p.2)
Q18	"We create a massive multitask test consisting of multiple-choice questions from various branches of knowledge. The test spans subjects in the humanities, social sciences, hard sciences, and other areas that are important for some people to learn. There are 57 tasks in total, which is also the number of Atari games (Bellemare et al., 2013), all of which are listed in Appendix B." (p.3)
Q28	"Finally, our history questions cover a wide range of time periods and geographical locations, including prehistory and other advanced subjects." (p.4)
Q33	"STEM subjects include physics, computer science, mathematics, and more." (p.4)
Q83	"By covering 57 subjects at varying levels of difficulty, the test assesses language understanding in greater breadth and depth than previous benchmarks." (p.8)
Q117	"Table 2: Summary of all 57 tasks." (p.15)
Q151	"Figure 44: A High School US History example." (p.21)
Q171	"Figure 68: A US Foreign Policy example." (p.26)
D2	"The song, and the movement that it was connected to, highlight which of the following developments in the broader society in the late 1800s?" (re: "The Hayseed", US Populist movement) — US-specific historical content with no South Asian relevance
D3	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — Entirely US constitutional law content
D4	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president. — US-specific legislative history
D5	In American government, the power to declare war rests with Congress. — US government structure; no relevance to South Asian governance or environmental policy
D6	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? — US Cold War policy — no South Asian agro-ecological relevance

D7	A truck driver was assigned to drive a huge tractor-trailer loaded with logs. The logs were being sold by a lumber company... the girl was struck by the tractor-trailer and was seriously injured. — US tort law scenario — US legal context
D8	An avid baseball fan learned that a local professional baseball club was conducting a baseball camp... forwarded the club a \$50 registration deposit — US contract law — US sports/cultural context
D16	The Scribbling-Machines have thrown thousands of your petitioners out of employ... Leeds Woolen Workers Petition, 1786 — Entirely European industrial history — no South Asian relevance
D17	What is tolerance?... Of all religions, the Christian ought doubtless to inspire the most tolerance... Voltaire, Letters on the English Nation, 1733 — European Enlightenment philosophy — no South Asian agricultural relevance
D20	Which zone contains low-income slums, ethnic ghettos, and general deterioration in Burgess's concentric **zone model of urban form***? — US urban planning model (Burgess model, Chicago School)
D25	Bill purchased a can of Sipep from the Ajax Minimart. After he finished drinking the Sipep, Bill noticed that the can contained dead insects stuck on the inside bottom of the can. In a strict product liability tort action against Ajax, Bill must prove... — US-style product liability tort — US legal culture

Information Gaps

- Approximately 20 of 57 configs were not directly sampled by the dataset analysis; rare South Asian agricultural items in unreviewed configs cannot be entirely ruled out, though structural sourcing makes them improbable.

Requires Expert Verification

- Whether any single MMLU item across the full 14,079-question test set addresses South Asian agro-ecology would require exhaustive scan of all 57 configs by a regional expert.

Remediation

Gap 1: MMLU's 57-subject taxonomy contains no agro-ecological, delta-ecology, or South Asian regional agricultural subjects — the precise constructs the deployment must measure.

Recommendation: Treat MMLU as a generic STEM/reasoning baseline only and pair it with a purpose-built South Asian agricultural knowledge benchmark organized by AEZ (haor, char-land, Deccan dry-land, coastal aquaculture) co-designed with BAU, BRRI, PJTSAU, and ANGRAU. Restrict MMLU reporting to a curated subset of culturally-neutral STEM configs (abstract algebra, college mathematics, college physics, formal logic, high school statistics).

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All MMLU content is sourced from US standardized tests (GRE, USMLE), undergraduate course materials, and Oxford University Press books [\[Q19\]](#), with Professional Law supplemented by Harvard Law Library case law [\[Q77\]](#) and history items drawing on English primary sources [\[Q140, Q141\]](#). The authorship team is entirely US-based [\[Q10\]](#) and funding is from US sources [\[Q87, Q88\]](#). Empirical sampling confirms US-centric content saturation: US tort law [\[DATASET-D7\]](#), US contract law [\[DATASET-D8\]](#), US Social Security [\[DATASET-D9\]](#), US birth control statistics [\[DATASET-D10\]](#), US Populist movement [\[DATASET-D2\]](#), MCAT references [\[DATASET-D39\]](#), and US measurement units [\[DATASET-D29\]](#). South Asian content is limited to three tangential references (India internet 2017, India vs. Congo university graduates, Jainism/samsara) and one colonially-framed Indian Uprising item [\[DATASET-D13, D40, D36, D18\]](#). No content from BRRI, BARI, BAU, ICAR, PJTSAU, or ANGRAU was identified, and gap searches confirmed no Bengali- or Telugu-language agricultural NLP corpus exists in the public landscape [\[WEB-22, WEB-6\]](#). The Mymensingh-dialect agricultural vocabulary, Telugu dry-land terminology, and cross-border water-policy content required for this deployment are absent by construction.

ID	Evidence
Q10	"Dan Hendrycks, UC Berkeley; Collin Burns, Columbia University; Steven Basart, UChicago; Andy Zou, UC Berkeley; Mantas Mazeika, UIUC; Dawn Song, UC Berkeley; Jacob Steinhardt, UC Berkeley" (p.1)
Q19	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q77	"We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy." (p.8)
Q87	"DH is supported by the NSF GRFP Fellowship and an Open Philanthropy Project Fellowship." (p.9)
Q88	"This research was also supported by the NSF Frontier Award 1804794." (p.9)
Q140	"This question refers to the following information." (p.19)
Q141	"English Parliament, Act of Supremacy, 1534" (p.19)
D2	"The song, and the movement that it was connected to, highlight which of the following developments in the broader society in the late 1800s?" (re: "The Hayseed", US Populist movement) — US-specific historical content with no South Asian relevance
D7	A truck driver was assigned to drive a huge tractor-trailer loaded with logs. The logs were being sold by a lumber company... the girl was struck by the tractor-trailer and was seriously injured. — US tort law scenario — US legal context
D8	An avid baseball fan learned that a local professional baseball club was conducting a baseball camp... forwarded the club a \$50 registration deposit — US contract law — US sports/cultural context
D9	The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context
D10	Which of the following is the most common method of birth control for married couples in the **U.S.** — Explicitly US demographic data
D13	As of 2017, what fraction of the population in **India** used the internet in the past three months? (A) 11% (B) 26% (C) 41% (D) 56% — South Asia-referenced factual question, but static 2017 statistic

D18	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters... British journalist William Howard Russell, My Indian Mutiny Diary, 1860 — British colonial framing of Indian history
D29	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? — US units (miles per hour) — not metric; minor cultural assumption
D36	In Jainism, what is the cycle that one must liberate oneself from? (A) Punya (B) Samsara — South Asian religion reference (Jainism) — one of very few South Asian cultural anchors in the dataset
D39	When preparing for the **MCAT exam** , a student begins studying electrochemical cells. — MCAT (US Medical College Admissions Test) reference — US academic system assumed
D40	At the time of independence, there were already hundreds of thousands of university graduates in **India** , but hardly any at all in **Congo** . — India referenced but as a factual datapoint about independence-era education, not agro-ecological
WEB-6	ICICC 2024 Telugu agricultural perplexity study explicitly notes lack of Telugu agricultural NLP resources (https://doi.org/10.1007/978-981-97-4152-6_38)
WEB-22	AI4Bharat IndicNLP Catalog — no Bangladeshi-Bengali or Mymensingh-dialect agricultural corpus listed (https://ai4bharat.github.io/indicnlp_catalog/)

Strengths

- A handful of global_facts and world_religions items reference South Asia, providing a minimal foothold of non-exclusively-Western reference [DATASET-D13, D36, D40]
- PDF/separate-page sourcing reduces verbatim memorization risk, preserving construct measurement integrity for the topics it does cover [Q110]

ID	Evidence
Q110	"We also note that most of our questions came from PDFs or websites where questions and answers are on separate pages." (p.14)
D13	As of 2017, what fraction of the population in **India** used the internet in the past three months? (A) 11% (B) 26% (C) 41% (D) 56% — South Asia-referenced factual question, but static 2017 statistic
D36	In Jainism, what is the cycle that one must liberate oneself from? (A) Punya (B) Samsara — South Asian religion reference (Jainism) — one of very few South Asian cultural anchors in the dataset
D40	At the time of independence, there were already hundreds of thousands of university graduates in **India** , but hardly any at all in **Congo** . — India referenced but as a factual datapoint about independence-era education, not agro-ecological

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — many MMLU items require US-specific cultural and institutional knowledge (US tort law, Social Security, MCAT system, US demographic statistics) [DATASET-D7, D9, D10, D11, D39]. Conversely, items requiring Mymensingh, Bangladeshi, or Telugu-region knowledge are absent.

IC-2: Culturally sensitive content (moral_scenarios anchored in 'US 2020 norms', US-centric birth control and aging facts) does not align with target deployment culture [DATASET-D1, D10, D11, D12]. Cross-border Bangladesh-India water-sharing topics — actively contested as of 2025 [WEB-1, WEB-2, WEB-4] — are not represented at all.

IC-3: Inputs requiring Western-specific knowledge are pervasive: US Constitutional structure [DATASET-D3, D5], US legislative history [DATASET-D4], Burgess Chicago-school urban model [DATASET-D20], UK educational sociology [DATASET-D26]. These do not transfer to South Asian agricultural contexts.

IC-4: INSUFFICIENT DOCUMENTATION on whether regional annotators reviewed any items; paper documents only US graduate/undergraduate collectors [Q19] and US MTurk baseline [Q22]. Full assessment would require recruiting Mymensingh, Telugu, and West Bengal agricultural experts to flag culturally sensitive instances.

IC-5: Combined with Q19's exclusively US sourcing and the empirical pattern across 175+ sampled examples, content validity for this deployment is severely compromised.

ID	Evidence
Q19	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q22	"Human-level accuracy on this test varies. Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test. Meanwhile, expert-level performance can be far higher. For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task. If we take the 95th percentile human test-taker accuracy for exams that build up our test, and if we make an educated guess when such information is unavailable, we then estimate that expert-level accuracy is approximately 89.8%." (p.3)
Q81	"Data may also become a bottleneck, as there is far less written about esoteric branches of knowledge than about everyday situations." (p.8)
D1	For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage
D3	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — Entirely US constitutional law content
D4	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president. — US-specific legislative history
D5	In American government, the power to declare war rests with Congress. — US government structure; no relevance to South Asian governance or environmental policy
D7	A truck driver was assigned to drive a huge tractor-trailer loaded with logs. The logs were being sold by a lumber company... the girl was struck by the tractor-trailer and was seriously injured. — US tort law scenario — US legal context
D9	The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context
D10	Which of the following is the most common method of birth control for married couples in the **U.S.** — Explicitly US demographic data
D11	From 1988 to 1990 among heterosexuals in the **US**, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US-specific behavioral survey data
D12	"For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms
D20	Which zone contains low-income slums, ethnic ghettos, and general deterioration in Burgess's concentric **zone model of urban form**? — US urban planning model (Burgess model, Chicago School)
D26	Smith & Tomlinson argued that school character far outweighed ethnic background in determining educational success — UK-centric sociological research
D39	When preparing for the **MCAT exam**, a student begins studying electrochemical cells. — MCAT (US Medical College Admissions Test) reference — US academic system assumed
WEB-1	Active Bangladesh-India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)
WEB-2	Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-4	Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)
WEB-6	ICICC 2024 Telugu agricultural perplexity study explicitly notes lack of Telugu agricultural NLP resources (https://doi.org/10.1007/978-981-97-4152-6_38)

WEB-22	AI4Bharat IndicNLP Catalog — no Bangladeshi-Bengali or Mymensingh-dialect agricultural corpus listed (https://ai4bharat.github.io/indicnlp_catalog/)
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Information Gaps

- Whether any unsampled config contains incidental Bangladesh- or Telugu-region agricultural content
- Whether annotators included any with South Asian regional expertise (paper does not document annotator demographics beyond US institutional affiliation)

Requires Expert Verification

- BRR/BAU agricultural extension experts could confirm absence of dialect-specific Mymensingh terminology
- Telugu agricultural linguistic experts (PJ TSAU/ANGRAU) could confirm absence of Deccan dry-land cropping vocabulary

Remediation

Gap 1: Content is entirely sourced from US/Anglophone institutions [Q19, Q77]; no Bengali- or Telugu-language agricultural content; cross-border policy context (Ganges/Teesta) absent.

Recommendation: Author a parallel evaluation set sourcing items from BRR/BAU technical bulletins, PJ TSAU/ANGRAU extension publications, and Joint Rivers Commission documents. Use the AgriPriceBD [WEB-23] LLM-assisted PDF parsing methodology to digitize Bengali agricultural reports. Build dialect-aware Mymensingh-Bengali corpus through BAU faculty elicitation (acknowledging this is unsearchable web-research territory).

ID	Evidence
Q19	“The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books.” (p.3)
Q77	“We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard’s Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy.” (p.8)
WEB-23	https://arxiv.org/abs/2604.06227

Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

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Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU is exclusively English-language, text-only, multiple-choice [Q8, Q21, Q45, Q46]. The HuggingFace metadata confirms monolingual English; every one of the 175+ examples sampled is English-only [DATASET-D2, D29, etc.]. The deployment elicitation explicitly requires Telugu, Bangladeshi Bengali (Mymensingh dialect), and Indian Bengali as query languages [elicitation A4, Q4]. Bengali script (বাংলা) and Telugu script (తెలుగు) are entirely unsupported. The paper's own authors acknowledge the limitation that 'many important concepts are conveyed mainly through other modalities' [Q64]. No transliteration, code-switching, or multilingual variant exists. Web research confirms no Bengali or Telugu agricultural-domain LLM evaluation infrastructure exists [WEB-5, WEB-17, WEB-18, WEB-25] and that LLMs perform meaningfully worse in Bengali than English [WEB-5]. This is an absolute, format-level mismatch that no in-format remediation can address. IF is a HIGH-priority dimension for this deployment, making this a fundamental validity violation.

ID	Evidence
Q4	"However, on every one of the 57 tasks, the best models still need substantial improvements before they can reach expert-level accuracy." (p.1)
Q8	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q21	"We collected 15908 questions in total, which we split into a few-shot development set, a validation set, and a test set. The few-shot development set has 5 questions per subject, the validation set may be used for selecting hyperparameters and is made of 1540 questions, and the test set has 14079 questions. Each subject contains 100 test examples at the minimum, which is longer than most exams designed to assess people." (p.3)
Q45	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with 'The following are multiple choice questions (with answers) about [subject].' For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with 'Answer: '. The model then produces probabilities for the tokens 'A,' 'B,' 'C,' and 'D,' and we treat the highest probability option as the prediction." (p.6)
Q46	"For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q64	"While text is capable of conveying an enormous number of concepts about the world, many important concepts are conveyed mainly through other modalities, such as images, audio, and physical interaction (Bisk et al., 2020). Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7)
D2	"The song, and the movement that it was connected to, highlight which of the following developments in the broader society in the late 1800s?" (re: "The Hayseed", US Populist movement) — US-specific historical content with no South Asian relevance
D29	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? — US units (miles per hour) — not metric; minor cultural assumption
WEB-5	arXiv 2507.23248 — LLMs perform worse in Bengali than English; smaller models especially degraded (https://arxiv.org/pdf/2507.23248)
WEB-17	AI4Bharat IndicBERT covers Bengali and Telugu but not agricultural domain (https://github.com/AI4Bharat/IndicBERT)
WEB-18	IndicTrans2 covers 22 Indian languages including Telugu but is a translation system, not knowledge benchmark (https://openreview.net/forum?id=vfT4YuzAYA)
WEB-25	IndicXTREME (ACL 2023) covers Bengali/Telugu for general NLU but no agricultural evaluation (https://aclanthology.org/2023.acl-long.693)

Strengths

- Standardized text-MCQ format is portable and computationally tractable for cross-model comparison [Q23, Q35]
- Format insensitivity findings for GPT-3 [Q100] mean that translated/adapted versions could in principle preserve relative model rankings if translation pipeline were added

ID	Evidence
Q23	"we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q35	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q100	"While different question formatting choices often lead to similar GPT-3 accuracies, we find that UnifiedQA is more sensitive." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution mismatch is total: source benchmark is English ASCII text [Q8, Q45]; target deployment requires Bengali script (Bangladeshi and Indian variants), Telugu script, and English. No regional language input is supported.

IF-2: Regional infrastructure does support text input — Bangladesh has 44.5–80% internet penetration [WEB-13, WEB-14] and mobile-first usage [WEB-15] — but this is irrelevant since MMLU's English-only constraint is the binding one, not regional capture capacity.

IF-3: Domain-specific form differences include: (a) Bengali/Telugu script Unicode handling (no uppercase/lowercase distinction), (b) absence of multimodal capacity (acknowledged limitation [Q64]) which excludes diagrams/maps relevant to delta hydrology and AEZ classification, (c) no code-switching support relevant to Bengali-English bilingual scientific register.

IF-4: Form mismatch is absolute and constitutes a complete external validity violation for the multilingual deployment requirement; no within-benchmark workaround exists.

ID	Evidence
Q8	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q45	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction." (p.6)
Q64	"While text is capable of conveying an enormous number of concepts about the world, many important concepts are conveyed mainly through other modalities, such as images, audio, and physical interaction (Bisk et al., 2020). Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7)
WEB-5	arXiv 2507.23248 — LLMs perform worse in Bengali than English; smaller models especially degraded (https://arxiv.org/pdf/2507.23248)
WEB-13	https://datareportal.com/reports/digital-2024-bangladesh
WEB-14	https://www.newagebd.net/post/telecom/252740/internet-penetration-rate-declines
WEB-15	https://en.wikipedia.org/wiki/Internet_in_Bangladesh
WEB-17	AI4Bharat IndicBERT covers Bengali and Telugu but not agricultural domain (https://github.com/AI4Bharat/IndicBERT)

Information Gaps

- Mymensingh-dialect-specific phonological/lexical variants in agricultural register are not documented in any retrievable NLP corpus and would require in-region linguist elicitation

Remediation

Gap 1: English-only text input precludes Bengali (both variants), Telugu, and any multilingual evaluation explicitly required by the deployment.

Recommendation: Do not use MMLU for any multilingual capability assessment. Build a multilingual evaluation pipeline using IndicXTREME [WEB-25] as the structural template and IndicTrans2 [WEB-18] for translation infrastructure, with native-script Bengali and Telugu items. Plan evaluation to detect language-mismatch failures (e.g., model responding in English to Bengali prompts).

ID	Evidence
WEB-18	IndicTrans2 covers 22 Indian languages including Telugu but is a translation system, not knowledge benchmark (https://openreview.net/forum?id=vfT4YuzAYA)
WEB-25	IndicXTREME (ACL 2023) covers Bengali/Telugu for general NLU but no agricultural evaluation (https://aclanthology.org/2023.acl-long.693)

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU applies a uniform four-class MCQ schema (A/B/C/D) across all 57 tasks [Q23, Q35], with no mechanism to encode legitimate regional divergence in correct answers. The deployment explicitly anticipates such divergence — particularly for Bangladesh–India water-sharing questions [elicitation A3] where active 2025 geopolitical context (Ganges Treaty renewal, unresolved Teesta dispute, Kushiya agreement) [WEB-1, WEB-2, WEB-4, WEB-21] would yield nationally divergent correct answers. The moral_scenarios config (500 items) explicitly anchors correctness to 'ordinary moral standards in the US as of 2020' as a constitutive framing across every item [DATASET-D1, D12], structurally precluding pluralism. The four-option forced-choice schema also cannot capture practice differences between delta flood-cycle planning and Deccan dry-land cropping. Models perform especially poorly on socially-important subjects (morality, law) [Q6, Q86] and on procedural/calculation tasks [Q14, Q72] — precisely the categories most relevant to agricultural decision-making. OO is HIGH-priority for this deployment, and the schema rigidity is a structural validity violation.

ID	Evidence
Q6	"Worse, they still have near-random accuracy on some socially important subjects such as morality and law." (p.1)
Q14	"The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality." (p.2)
Q23	"we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q35	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q72	"Models also have difficulty performing calculations, so much so that they exhibit poor performance on Elementary Mathematics and many other STEM subjects with "plug and chug" problems." (p.8)
Q86	"Worryingly, models also perform especially poorly on socially relevant subjects including morality and law." (p.8)
D1	For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage
D12	"For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms
WEB-1	Active Bangladesh–India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)
WEB-2	Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-4	Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)
WEB-21	Kushiya River Agreement — recent bilateral agreement not reflected in 2021 benchmark labels (https://www.waterdiplomat.org/story/2025/08/teesta-river-politics-and-benefit-sharing-getting-yes-without-grand-bargain)

Strengths

- Four-option MCQ schema is simple and reproducible for cross-model accuracy comparison [Q23, Q35]

- Difficulty levels (Elementary/High School/College/Professional) provide some stratification useful for capacity profiling [Q20]

ID	Evidence
Q20	"Some tasks cover a subject, like psychology, but at a specific level of difficulty, such as "Elementary," "High School," "College," or "Professional."" (p.3)
Q23	"we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q35	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The MCQ label space (A/B/C/D) [Q23] is generic and content-agnostic; it is not regionally relevant or irrelevant per se, but it cannot represent the categorical decision boundaries needed for regional agricultural reasoning (e.g., 'optimal under delta flood regime' vs. 'optimal under Deccan dry-land regime' as parallel correct answers).

OO-2: Missing categories include: regional-pluralism markers (which national policy regime applies), agro-ecological zone qualifiers, and any output that would let a model express 'this depends on whether the question is from a Bangladeshi or Indian regulatory perspective' [WEB-1, WEB-2 — active treaty divergence].

OO-3: The moral_scenarios subtask explicitly encodes US 2020 norms as the constitutive correctness criterion [DATASET-D1, D12], embedding non-regional values directly into the output ontology. Human_sexuality and human_aging similarly embed US institutional facts as universally correct outputs [DATASET-D9, D10, D11].

OO-4: Significant misalignment exists; stakeholder-driven taxonomy redesign would require either (a) adding parallel-answer schema for regionally-divergent items or (b) excluding the affected subtasks. The paper itself notes models 'are notably poor at modeling human (dis)approval' [Q71], compounding OO concerns on socially-situated items.

OO-5: The combination of forced single-correct-answer MCQ + US-anchored normative subtasks + procedural-calculation weakness [Q72] across calculation-heavy tasks (Elementary Mathematics RMS calibration error 19.4% [Q62]) creates structural validity issues for an agricultural decision-support evaluation.

ID	Evidence
Q6	"Worse, they still have near-random accuracy on some socially important subjects such as morality and law." (p.1)
Q14	"The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality." (p.2)
Q23	"we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q62	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%." (p.7)
Q71	"They are notably poor at modeling human (dis)approval, as evident by the low performance on the Professional Law and Moral Scenarios tasks." (p.8)
Q72	"Models also have difficulty performing calculations, so much so that they exhibit poor performance on Elementary Mathematics and many other STEM subjects with "plug and chug" problems." (p.8)
Q86	"Worryingly, models also perform especially poorly on socially relevant subjects including morality and law." (p.8)

D1	For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage
D9	The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context
D10	Which of the following is the most common method of birth control for married couples in the **U.S.** — Explicitly US demographic data
D11	From 1988 to 1990 among heterosexuals in the **US**, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US-specific behavioral survey data
D12	"For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms
WEB-1	Active Bangladesh–India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)
WEB-2	Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-4	Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)

Information Gaps

- Whether any individual MMLU item explicitly flags regional divergence in its answer key (none observed in sample)

Requires Expert Verification

- Bangladeshi and Indian water-policy experts would be needed to systematically identify items whose correct answer differs by national perspective

Remediation

Gap 1: Four-option MCQ schema cannot encode legitimate regional divergence in correct answers (especially cross-border water/agricultural policy); moral_scenarios encodes US 2020 norms structurally [DATASET-D1, D12].

Recommendation: Exclude moral_scenarios, human_sexuality, human_aging, and US-jurisdiction-specific configs from any deployment-relevant scoring. For the supplementary regional benchmark, design schema permitting parallel-correct-answer annotation flagged by jurisdiction (Bangladesh / India / state-level) for water-policy and agricultural-policy items.

ID	Evidence
D1	For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage
D12	"For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The paper is silent on annotator demographics, selection criteria, regional or disciplinary backgrounds, or inter-annotator agreement [\[Q19\]](#). The only documented human reference is US Amazon Mechanical Turk workers (34.5% accuracy) [\[Q22\]](#), with expert-level estimated from 'real-world test-taker human accuracy at the 95th percentile' on US standardized exams. The HuggingFace dataset card tags annotations_creators as 'no-annotation' [\[DATASET Concern 8\]](#). The authorship team is entirely US-based [\[Q10\]](#) and content is sourced from US/Anglophone institutions [\[Q19, Q77\]](#). There is no evidence — and no plausible mechanism — for South Asian agronomic, Mymensingh-dialect, or Telugu-region expert validation of any answer key. The user explicitly confirmed [\[elicitation A3\]](#) that cross-border water/agricultural questions could yield divergent correct answers and that practice differences between regions are not reflected in answer keys. Empirical sampling confirmed multiple items where US-specific facts (Social Security age increases, US birth control prevalence, US 2020 moral norms) are encoded as universally correct [\[DATASET-D1, D9, D10, D11, D12\]](#). OC is HIGH-priority and represents a fundamental convergent and external validity violation.

ID	Evidence
Q10	"Dan Hendrycks, UC Berkeley; Collin Burns, Columbia University; Steven Basart, UChicago; Andy Zou, UC Berkeley; Mantas Mazeika, UIUC; Dawn Song, UC Berkeley; Jacob Steinhardt, UC Berkeley" (p.1)
Q19	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q22	"Human-level accuracy on this test varies. Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test. Meanwhile, expert-level performance can be far higher. For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task. If we take the 95th percentile human test-taker accuracy for exams that build up our test, and if we make an educated guess when such information is unavailable, we then estimate that expert-level accuracy is approximately 89.8%." (p.3)
Q77	"We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy." (p.8)
D1	For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage
D9	The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context
D10	Which of the following is the most common method of birth control for married couples in the **U.S.** — Explicitly US demographic data
D11	From 1988 to 1990 among heterosexuals in the **US**, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US-specific behavioral survey data
D12	"For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms

Strengths

- Authors commit to publishing question sources to enable downstream auditing [\[Q112\]](#)

- Memorization analysis suggests labels are not artifacts of pretraining contamination [Q107, Q108], so label correctness is at least an honest measurement of US-anchored ground truth

ID	Evidence
Q107	"However, in Figure 13, we see that accuracy and question entropy are not positively correlated, suggesting that the test's low-entropy questions do not correspond to memorized (and thereby correctly predicted) answers." (p.14)
Q108	"This suggests that our exact questions were not memorized." (p.14)
Q112	"To reduce the probability that future models encounter exact questions during test-time, we will provide a list of question sources." (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels reflect US/Anglophone academic standards [Q19, Q22], not Bangladeshi or Telugu-region stakeholder perspectives. The user [elicitation A3] confirms they would not be considered authoritative for cross-border or Bangladeshi-context agricultural questions.

OC-2: Substantial disagreement is expected for: (a) moral_scenarios items (US 2020 framing) [DATASET-D1, D12]; (b) any cross-border water-policy item [WEB-1, WEB-2, WEB-4]; (c) human_aging/human_sexuality US institutional facts [DATASET-D9, D10, D11]; (d) potentially historical framings such as the Cawnpore item [DATASET-D18].

OC-3: INSUFFICIENT DOCUMENTATION — paper provides no Datasheet/Data Statement; only documents that questions were 'manually collected by graduate and undergraduate students' [Q19] without demographic breakdown. HF metadata indicates 'no-annotation' [DATASET Concern 8].

OC-4: Re-annotation by representative regional annotator pool (BAU, BRRI, PJTSAU, ANGRAU agronomists; Bangladeshi and Indian water-policy experts) would be required for any deployment-relevant subset; the paper offers no such pool.

OC-5: Aggregation methods are not described; the single-correct-answer MCQ format inherently erases minority perspectives [Q23], which is particularly concerning for cross-border and culturally-situated questions.

OC-6: The combined absence of regional annotator validation, the explicit US-normative framing of large subtasks, and the user-confirmed expectation of divergent correct answers constitute serious convergent and external validity violations.

ID	Evidence
Q19	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q22	"Human-level accuracy on this test varies. Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test. Meanwhile, expert-level performance can be far higher. For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task. If we take the 95th percentile human test-taker accuracy for exams that build up our test, and if we make an educated guess when such information is unavailable, we then estimate that expert-level accuracy is approximately 89.8%." (p.3)
Q23	"we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)

D1	For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage
D9	The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context
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D12	"For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms
D18	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters... British journalist William Howard Russell, My Indian Mutiny Diary, 1860 — British colonial framing of Indian history
WEB-1	Active Bangladesh-India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)
WEB-2	Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-4	Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)
WEB-21	Kushiyara River Agreement — recent bilateral agreement not reflected in 2021 benchmark labels (https://www.waterdiplomat.org/story/2025/08/teesta-river-politics-and-benefit-sharing-getting-yes-without-grand-bargain)

Information Gaps

- Annotator demographic information is entirely absent from documentation
- No quality-assurance or inter-annotator agreement statistics reported

Requires Expert Verification

- Specific items where Bangladeshi vs Indian agronomic standards would diverge would require BAU/PJTSAU/ANGRAU expert review
- Cross-border water-policy item answer keys would require Joint Rivers Commission expert adjudication

Remediation

Gap 1: No South Asian annotators were involved in label creation; user confirmed cross-border questions would yield divergent correct answers; HF metadata tags annotations as 'no-annotation'.

Recommendation: For any subset of MMLU items used in deployment evaluation, conduct a relabeling pass with at least three annotators per item drawn from a pool spanning Bangladesh (BAU, BRRI), Telangana (PJTSAU), and Andhra Pradesh (ANGRAU). Track inter-annotator disagreement rates and exclude items where regional disagreement exceeds threshold. Document annotator demographics in a Datasheet.

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU evaluates only text-based MCQ classification accuracy [Q23, Q35] via predicted A/B/C/D tokens [Q45]. The metric (accuracy) and calibration measures (RMS calibration error, accuracy-confidence gap) [Q60, Q62] partially align with the deployment's stated need for cross-model comparison. However, the format cannot evaluate open-ended agronomic reasoning, does not penalize language mismatches (e.g., model responding in English to a Bengali prompt), and does not support regional-language outputs. The user did not flag output form as a primary concern (MODERATE priority), and MCQ accuracy is at least usable as a baseline. But the format precludes evaluating practice-difference reasoning, generative explanations of flood-cycle decisions, or Bengali/Telugu-language extension communication — all relevant for downstream farmer/agronomist applications. Combined with the 'esoteric knowledge' data bottleneck [Q81] amplified for South Asian agricultural content, output form is partially aligned but insufficient for deployment-validity evaluation.

ID	Evidence
Q23	"we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q35	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q45	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction." (p.6)
Q60	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)
Q62	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%." (p.7)
Q81	"Data may also become a bottleneck, as there is far less written about esoteric branches of knowledge than about everyday situations." (p.8)

Strengths

- Single accuracy metric is reproducible and supports cross-model comparison [Q35, Q42]
- Calibration analysis (RMS calibration error, accuracy-confidence gap) provides useful uncertainty diagnostics for downstream deployment risk assessment [Q60, Q61, Q62, Q106]
- Per-task and per-grouping reporting [Q52, Q92] would in principle let a user isolate STEM-reasoning subscores even while ignoring US-content subscores

ID	Evidence
Q35	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q42	"All values are percentages." (p.5)
Q52	"Figure 6 shows the accuracy of GPT-3 (few-shot) and UnifiedQA for all 57 tasks. It shows the both models are below expert-level performance for all tasks, with GPT-3's accuracy ranging from 69% for US Foreign Policy to 26% for College Chemistry." (p.6)
Q60	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)

Q61	"GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects." (p.7)
Q62	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%." (p.7)
Q92	"On the left are GPT-3 few shot accuracies for all of the 57 tasks. On the right are UnifiedQA transfer accuracies for all of the 57 tasks. For both models, capabilities are lopsided." (p.11)
Q106	"While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is $r = 0.81$, compared to $r = 0.63$ in the zero-shot setting." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (single letter token) [Q45] does not match deployment needs — environmental science applications often require open-ended explanation, calculation, or recommendation. The MCQ format aligns only with the cross-model comparison goal.

OF-2: Text-to-speech is not relevant to MMLU's evaluation metric; not applicable to this benchmark form.

OF-3: Literacy considerations: Bangladesh national literacy is 77.9% [WEB-7], functional literacy 62.92% [WEB-8]; Telangana 66.54% and Andhra Pradesh 67.02% [WEB-9, WEB-11]. The downstream farmer/agronomist sub-population would not directly interact with an English-MCQ output anyway, but for any extension-tool evaluation, output form would need to support Bengali/Telugu generative output — which MMLU does not assess.

OF-4: Form mismatches are real but partial; MCQ accuracy can serve as a baseline benchmark while requiring substantial supplementation (open-ended generation, multilingual generation evaluation) for deployment-level validity.

ID	Evidence
Q35	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q45	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction." (p.6)
Q81	"Data may also become a bottleneck, as there is far less written about esoteric branches of knowledge than about everyday situations." (p.8)
WEB-7	Bangladesh literacy 77.9% (BBS 2024) (https://www.observerbd.com/news/542998)
WEB-8	Bangladesh functional literacy 62.92% (BBS LAS 2023) (https://en.prothomalo.com/bangladesh/c7axno0vhh)
WEB-9	Telangana literacy 66.54% (Census 2011) (https://testbook.com/question-answer/what-is-the-literacy-rate-of-telangana--5efeb788fbc9b00d102b5fda)
WEB-11	Andhra Pradesh literacy 67.02% (Census 2011) (https://www.census2011.co.in/census/state/andhra+pradesh.html)

Information Gaps

- Whether deployment will require open-ended generation or remain MCQ-style was not fully specified in elicitation

Remediation

Gap 1: MCQ accuracy cannot capture open-ended agronomic reasoning needed for downstream farmer/agronomist applications, and does not penalize language-mismatch failures.

Recommendation: Supplement MMLU MCQ accuracy with open-ended generative evaluation in Bengali and Telugu, evaluated by regional experts using rubrics that explicitly penalize language drift (English response to Bengali prompt) and reward dialect-appropriate register for Mymensingh-Bengali agricultural extension communication.